THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** sour	Acid.
2	**(b)** electricity	Electric energy.
3	**(b)** metal	Metal.
4	**(d)** physical change	Physical.
5	**(a)** hydrochloric acid	HCl.
6	**(a)** copper	Good conductor.
7	**(b)** copper	Metal.
8	**(a)** melting of ice	Melting.
9	**(a)** vinegar	Vinegar.
10	**(b)** cap	Metal cap.
11	**(b)** metal oxide	Oxide.
12	**(b)** carbon dioxide	Fizz.
13	**(d)** soapy	Bases are bitter.
14	**(b)** two or more cells	Combination of cells.
15	**(a)** non-metal	Non-metal.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A sour-tasting substance (1); Vinegar (1).	1+1
17	A source of electricity (1); It has chemicals (1).	1+1
18	Iron (1); Copper (1).	1+1
19	No new substance (1); E.g. melting (1).	1+1
20	Turns red (1); Shows acid (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Lemon (1); Vinegar (1); Curd (1).	1+1+1
22	Cell (1); Bulb (1); Switch (1).	1+1+1
23	Lustre (1); Malleable (1); Good conductors (1).	1+1+1
24	Melting of ice (1); Boiling of water (1); Cutting paper (1).	1+1+1
	OR: Citric: lemon (1); Lactic: curd (1); Acetic: vinegar (1).	1+1+1
25	Cell: long and short lines (1); Bulb: circle with a cross (1); Switch: a gap with a lever (1).	1+1+1
	OR: Good conductor (1); Ductile (1); Long lasting (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Acid sour, base bitter (1); Acid: blue litmus red (1); Base: red litmus blue (1); Base feels soapy (1); Examples differ (1).	1+1+1+1+1
	OR: Take a cell, a bulb and wires (1); Join one end of the bulb to + of the cell (1); Join the other end to - of the cell (1); The circuit is complete (1); The bulb glows (1).	1+1+1+1+1
27	Lustre (1); Malleability (1); Ductility (1); Conductivity (1); Sonority (1).	1+1+1+1+1
	OR: Melting (1); Boiling (1); Freezing (1); Tearing paper (1); Dissolving sugar (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** chemical change	Chemical.
2	**(d)** sulphuric acid	It has sulphur.
3	**(b)** plastic	Insulation.
4	**(a)** rubber	Non-metal.
5	**(d)** burning of paper	Burning.
6	**(c)** soap	Soap.
7	**(b)** flat base	Flat.
8	**(d)** non-metal oxide	Oxide.
9	**(a)** carbon dioxide	CO_2.
10	**(a)** acidic	Acetic acid.
11	**(a)** filament	Thin wire.
12	**(d)** mercury	Mercury.
13	**(b)** chemical change	Chemical.
14	**(a)** nitric acid	Nitric.
15	**(c)** safety	Protects circuits.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Copper (1); Iron (1).	1+1
17	Drawn into wires (1); Copper wire (1).	1+1
18	Iron oxide (1); Reddish-brown (1).	1+1
19	Citric acid (1); Lactic acid (1).	1+1
20	A broken path (1); No current (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Rain (1); Seasons (1); Day and night (1).	1+1+1
22	Turns red (1); Soap is basic (1); Yellow in acid (1).	1+1+1
23	To turn on (1); To turn off (1); To save energy (1).	1+1+1
24	Mercury is liquid (1); Graphite conducts (1); Sodium is soft (1).	1+1+1
	OR: Wax remains wax (1); Solidifies again (1); No new substance (1).	1+1+1
25	Soap (1); Baking soda (1); Lime water (1).	1+1+1
	OR: Copper (1); Iron (1); Aluminium (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Litmus: red, blue (1); Turmeric: yellow, red (1); China rose: pink, green (1); Acid first (1); Base second (1).	1+1+1+1+1
	OR: Light (1); Heat (1); Fan (1); TV (1); Cooking (1).	1+1+1+1+1
27	Burn sulphur (1); Gas forms (1); Dissolve in water (1); Acid forms (1); Blue litmus turns red (1).	1+1+1+1+1
	OR: Add alum to hot water (1); Stir (1); Filter (1); Cool (1); Crystals form (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** iron	Metal.
2	**(a)** reddish-brown	Colour of rust.
3	**(a)** water	Water.
4	**(c)** connections using symbols	Standard drawing.
5	**(c)** white flame	Dazzling white.
6	**(a)** rusting	Slow.
7	**(d)** basic	Base.
8	**(b)** open or close a circuit	On/off.
9	**(c)** shiny	Lustre.
10	**(c)** physical change	Physical.
11	**(a)** sodium hydroxide	A base.
12	**(a)** shock	Dangerous.
13	**(c)** sulphur	Brittle.
14	**(b)** water	Moisture.
15	**(a)** soap solution	Base.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sulphur (1); Rubber (1).	1+1
17	Can be undone (1); E.g. freezing (1).	1+1
18	A bitter, soapy substance (1); Soap (1).	1+1
19	Combination of cells (1); Gives more voltage (1).	1+1
20	Sulphur (1); Oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Dull (1); Brittle (1); Poor conductors (1).	1+1+1
22	Burning (1); Rusting (1); Cooking (1).	1+1+1
23	Soap (1); Antacid (1); Cleaning (1).	1+1+1
24	Connect a cell (1); Close the switch (1); Use wires (1).	1+1+1
	OR: Malleable (1); Light (1); Non-toxic (1).	1+1+1
25	Cooking (1); Making a house (1); Making ice (1).	1+1+1
	OR: Dark pink (1); Acid (1); Green in base (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Take red and blue litmus (1); Dip in the solution (1); See the colour change (1); Acid turns blue to red (1); Base turns red to blue (1).	1+1+1+1+1
	OR: Copper (1); Iron (1); Aluminium (1); Rubber (1); Plastic (1).	1+1+1+1+1
27	Lustre (1); Hardness (1); Brittleness (1); Conduction (1); Sound (1).	1+1+1+1+1
	OR: Burning (1); Rusting (1); Cooking (1); Digestion (1); Fermentation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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